

GASPE, QC, Canada

WMO: 711880

Lat: 48.7769N

Lon: 64.4781W

Elev: 112

StdP: 14.64

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.8	-7.3	-22.4	1.6	-9.8	-17.6	2.1	-6.0	27.8	23.2	23.8	21.7	5.0	270	0.645

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.4	82.4	67.4	78.7	65.9	75.4	64.2	69.8	78.7	67.8	75.8	66.0	73.1	10.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.5	98.1	74.9	64.6	91.6	72.2	63.0	86.3	70.2	34.0	78.9	32.3	75.9	30.8	73.1	77.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.1	18.9	16.8	-19.0	90.3	4.3	3.1	-22.0	92.6	-24.5	94.4	-26.9	96.1	-30.0	98.4		
(5)			-19.4	73.8	4.2	1.9	-22.4	75.1	-24.9	76.3	-27.2	77.3	-30.3	78.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.8	13.1	13.8	22.9	34.6	45.9	56.3	63.4	62.2	54.1	43.3	32.6	21.4		
	DBStd	19.16	10.71	10.66	9.78	6.38	6.51	6.97	5.39	5.31	6.66	6.80	8.22	9.11		
	HDD50	5317	1144	1015	840	462	166	17	0	0	30	231	524	888		
	HDD65	9672	1609	1435	1305	912	594	275	93	119	332	673	972	1353		
	CDD50	1219	0	0	0	1	38	205	417	378	154	23	3	0		
	CDD65	98	0	0	0	0	1	14	44	33	6	0	0	0		
	CDH74	1027	0	0	0	0	37	194	408	313	70	5	0	0		
	CDH80	225	0	0	0	0	8	50	87	66	14	0	0	0		
	WSAvg	6.9	7.5	7.5	7.8	7.5	6.7	6.4	5.7	5.7	6.1	6.9	7.1	7.7		
	PrecAvg	45.0	3.3	3.3	3.9	2.8	3.5	3.4	4.0	4.0	3.5	4.8	3.8	4.8		
(15) Precipitation	PrecMax	58.9	5.0	5.8	10.3	6.3	6.8	7.1	6.4	8.8	7.7	11.1	6.7	16.0		
	PrecMin	31.3	0.5	0.9	1.0	0.9	1.1	0.9	1.2	0.8	0.4	1.1	1.6	1.4		
	PrecStd	6.1	1.1	1.3	2.0	1.6	1.5	1.3	1.5	1.8	1.7	2.0	1.3	2.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.1	44.3	49.0	60.7	79.7	86.2	87.1	86.5	81.6	72.3	61.1	46.8		
		MCWB	44.5	41.9	42.3	47.6	61.0	68.9	69.6	69.4	67.4	60.1	57.0	45.1		
	2%	DB	37.8	37.4	43.1	53.7	70.4	79.3	82.1	81.0	74.6	63.9	54.4	40.7		
		MCWB	35.6	35.0	37.7	45.1	56.4	64.2	68.0	67.4	64.0	56.6	50.2	38.5		
	5%	DB	33.5	33.6	39.6	49.0	64.3	74.5	78.4	77.2	70.6	59.3	48.6	37.2		
		MCWB	31.5	31.4	34.5	41.3	52.9	62.4	66.7	65.5	62.0	52.8	44.8	35.1		
	10%	DB	29.4	30.3	36.1	45.2	59.6	70.4	75.2	73.9	66.9	55.0	44.3	34.0		
		MCWB	27.5	28.3	32.5	38.9	50.0	59.9	64.6	63.9	59.2	49.9	41.0	32.4		
	0.4%	WB	44.8	42.4	43.6	52.2	63.5	70.3	72.9	71.7	69.5	61.7	57.5	45.2		
		MCDB	46.0	44.0	47.8	57.6	74.9	81.5	83.0	82.1	77.8	68.6	60.8	46.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.8	35.4	38.3	45.9	58.0	66.8	70.3	69.2	65.7	57.3	50.8	39.0		
		MCDB	37.6	36.7	42.2	51.7	67.7	75.7	79.1	77.9	72.9	63.0	53.5	40.0		
	5%	WB	31.9	31.7	35.3	42.4	54.5	63.9	68.2	67.2	63.2	53.8	45.4	35.5		
		MCDB	33.2	33.3	38.6	48.2	62.6	72.2	75.8	74.9	69.2	58.5	47.8	36.9		
	10%	WB	27.7	28.3	32.9	39.4	50.9	61.1	66.2	65.3	60.3	50.3	41.4	32.8		
		MCDB	29.3	30.2	35.6	44.0	58.0	68.8	73.0	72.0	65.5	54.1	43.7	33.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.8	18.8	17.5	15.8	20.3	21.0	20.4	20.8	20.5	16.6	13.3	13.6		
		MCDBR	21.3	18.9	20.5	23.6	31.3	29.2	26.4	26.7	26.6	24.3	19.0	16.2		
		MCWBR	20.5	17.2	15.8	16.0	18.7	16.2	14.4	14.7	16.9	17.2	16.4	15.3		
	5% WB	MCDBR	21.2	17.1	18.4	21.7	27.7	25.8	22.7	22.8	23.4	22.1	17.8	15.6		
		MCWBR	20.6	16.2	15.0	15.9	17.5	15.4	13.5	13.7	16.0	16.7	16.4	15.1		
(40) Clear-Sky Solar Irradiance	taub		0.275	0.279	0.310	0.351	0.363	0.382	0.412	0.383	0.357	0.338	0.305	0.282		
	taud		2.357	2.301	2.298	2.347	2.414	2.364	2.331	2.404	2.462	2.492	2.497	2.403		
	Ebn at Noon		252	279	286	285	285	279	268	272	268	255	241	234		
	Edh at Noon		21	28	34	36	35	37	38	34	29	24	19	18		
(44) All-Sky Solar Radiation	RadAvg		369	637	1036	1390	1627	1773	1704	1493	1135	697	365	265		
	RadStd		33	60	87	116	119	144	150	87	88	51	43	25		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	+2.11	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (15 neighbors)		N/A	N/A	+2.59	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page